

Public datasets for publishing the reference-diff + VLM approach

The goal of this document is to settle the question: which public benchmarks do we evaluate on, so that a paper about `vlm_05_reference_diff.py` is not rejected on the grounds that it was only ever tested on our own data from DPO tram.

The problem

What ARSI does is specific. Fixed CCTV cameras inside tram interiors, looking for changes that persist: graffiti, vandalism, forgotten objects. The method is to subtract an inspection frame from a clean reference frame of the same camera, and hand each region that differs to a vision-language model for judgement.

No public dataset covers that combination. In-vehicle, fixed camera, and ground truth for graffiti or abandoned objects simply do not exist together in anything published.

So the plan is not to go looking for one dataset that does everything. The pipeline is really three separate problems stacked on top of each other: detecting that something in the image changed, deciding whether that change is persistent and abnormal, and interpreting what it actually is. Each of those gets evaluated on the benchmark where it is meaningful, and the Ostrava tram data becomes the deployment section rather than the only evidence in the paper.

I don't think having our own dataset is what gets a paper rejected. Having our own dataset and nothing else is.

Where the compact-VLM paper stands

Borodin et al., *Benchmarking Compact VLMs for Clip-Level Surveillance Anomaly Detection Under Weak Supervision*, J. Imaging 11(11):400, November 2025. Everything they report is on UCF-Crime dataset.

UCF-Crime is an anomaly dataset: 1900 untrimmed videos, 128 hours, 13 anomaly classes, with labels given at the level of the whole video rather than per frame (this is what "weak supervision" means here). The reason it cannot carry our main result is not that the anomalies are the wrong kind. It is that every video comes from a different scene and a different camera, and there is no clean footage of that same camera to use as a reference. Our differencing stage has nothing to subtract from.

The evaluation plan

QUESTION WE WANT TO ANSWER	BENCHMARK	WHAT WE MEASURE
Does it detect changes correctly?	CDnet 2014	precision, recall, F1 and false-positive rate, computed per pixel
How badly do camera shake, shadows, changing light and moving backgrounds hurt it?	CDnet 2014, broken down by category	how much each category degrades the score
Do persistent changes actually turn into abandoned-object detections?	i-LIDS AVSS 2007, PETS2006 and PETS2007, ABODA, using the Luna et al. protocol	precision, recall and F1 at the level of events rather than pixels
How does our VLM stage compare with recent work?	UCF-Crime, following the Borodin protocol	accuracy, precision, recall, F1, ROC-AUC, latency

QUESTION WE WANT TO ANSWER	BENCHMARK	WHAT WE MEASURE
Does the system meet the actual tram requirement? Graffiti, damaged seats, stains, phones, bottles	our Ostrava data	precision, recall and F1 per class, plus ablations

Read as a paper outline, that becomes:

1. CDnet 2014	evaluate the change-proposal mechanism against established change-detection methods
2. PETS / AVSS	show that persistent changes translate into real abandoned-object detections
3. UCF-Crime	compare the VLM stage against recent compact-VLM work
4. Ostrava	evaluate the complete system in the deployment domain, including anomaly classes no public benchmark covers

Notes on each dataset

CDnet 2014

Fifty-three sequences across 11 categories, all from static cameras, with ground truth given for every pixel of every frame. The category called `intermittentObjectMotion` is defined by its authors as "videos containing background objects moving away, abandoned objects and objects stopping for a short while and then moving away", which is our case almost word for word. Its six sequences:

SEQUENCE	FRAMES
abandonedBox	4 500
parking	2 500
streetLight	3 200
sofa	2 750
tramstop	3 200
winterDriveway	2 500

The other categories let us stress the differencing stage exactly where we already suspect it is weak: `shadow`, `dynamicBackground`, `cameraJitter`, `badWeather`, `nightVideos` and `lowFramerate`.

Now the caveat. **CDnet is not an anomaly-detection benchmark.** Its ground truth marks each pixel as one of: static (0), hard shadow (50), outside the region of interest (85), unknown motion such as motion blur or semi-transparency (170), or moving foreground (255). Nothing in there knows that a bottle has been forgotten, that graffiti is a problem, or that a seat has been slashed. So the claim we are entitled to make is:

Correct: "our change-detection stage reaches $F1 = 0.84$ on `intermittentObjectMotion`"

Not correct: "our anomaly detector reaches $F1 = 0.84$ "

Within that limit it is very defensible, and it buys us the one ablation we are currently missing. **CDnet lets us evaluate `vlm_05_reference_diff.py` without the VLM in the loop.** When the full pipeline gets something wrong, we will finally be able to say whether the differencing stage failed to propose the region, or whether it proposed it correctly and the VLM misjudged it. [LORA PLAN.md](#) already claims localization is not the bottleneck, based on 45 out of 45 ground-truth objects found on our own data; CDnet is what lets us make that claim on data someone else built.

- Paper: https://openaccess.thecvf.com/content_cvpr_workshops_2014/W12/papers/Wang_CDnet_2014_An_2014_CVPR_paper.pdf
- Site: <https://changedetection.net/dataset2014/> (there is also a Kaggle mirror)

Abandoned objects: use the Luna protocol

i-LIDS AVSS 2007 has an explicit Abandoned Baggage challenge filmed in a railway station. Its official definition is useful to us on its own: the owner must have left the area and not come back for 60 seconds, and the object types covered are bottle, can, suitcase, newspaper, paper, rucksack and sports bag. That is close to what ARSI means by a forgotten object, persistence criterion included.

PETS2006 is seven scenarios of abandoned luggage in a station, filmed simultaneously from four viewpoints at 25 frames per second, with ground truth and increasing levels of difficulty.

ABODA is 11 CCTV sequences of unattended objects, indoors and outdoors, day and night, with strong lighting changes. Semantically it is exactly our class, but 11 sequences is too little to carry a main results table, and the licence is not clearly stated anywhere. Use it for qualitative checks.

The useful find here is a paper rather than a dataset. Luna et al., "Abandoned Object Detection in Video-Surveillance: Survey and Comparison", *Sensors* 18(12):4290, 2018, builds a *common evaluation protocol* across AVSS AB 2007, PETS2006, PETS2007, ABODA and VISOR. It unifies the temporal annotations for abandoned objects and provides baseline results for each stage of the usual pipeline (Mixture of Gaussians, K-nearest neighbours, PAWCS, and others, all background-subtraction methods). Adopting it is much stronger than assembling our own evaluation.

The original hosting for AVSS 2007 is unreliable, but mirrors exist and the authors' own package is reportedly still available.

- <https://mdpi.com/1424-8220/18/12/4290/htm>
- AVSS 2007 challenge definition: <https://www.eecs.qmul.ac.uk/~andrea/dwnld/avss2007/>

Bus Violence: in-vehicle, but not usable as a benchmark for us

Ciampi et al., *Sensors* 22(21):8345, 2022, from CNR-ISTI in Pisa. It contains 1400 clips, 700 with a violent action and 700 without, filmed by three cameras inside a moving bus: two in the corners at 960×540 and one fisheye in the middle at 1280×960, all at 25 frames per second. The whole thing is about 449 MB, hosted on Zenodo as record 7044203.

Two facts kill the obvious framing of it:

- 01 **The clips are very short: between 16 and 48 frames at 25 frames per second, so 0.6 to 1.9 seconds each.** This is not a test of stability over ten minutes of a tram moving from sunlight into shade, and it must not be described as one.
 - 02 **"NoViolence" does not mean "normal" in our sense.** The label only asserts that no violent action takes place. It says nothing about whether a bag was left behind, whether there is a stain, whether something is damaged, or whether there is graffiti on the wall. Those 700 clips are not 700 verified negatives for our task and cannot be counted as such.
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC9658862/> and <https://zenodo.org/records/7044203>

UCF-Crime: a separate compatibility experiment

This is not our main benchmark. To end up in the same table as the compact vision-language models, it is not enough to use the same dataset; the whole protocol has to be reproduced. Verified from the paper:

Split	the official one, no filtering, no separate validation subset
Training	1610 clips (800 normal, 810 abnormal)
Test	290 clips (150 normal, 140 abnormal)
Sampling	videos under 60 s: 0.5 frames per second across the whole video videos 60 s or more: 32 frames spread evenly across the timeline no audio, no optical flow
Decision	one binary label per clip, inherited from the source video; no temporal localization. Output constrained to 0 or 1, except in chain-of-thought mode where the reasoning precedes the digit
Models	2-4B: Gemma-3-4B, InternVL3-2B, Qwen2.5-VL-3B 7-8B: InternVL3-8B, Qwen2.5-VL-7B, Video-LLaVA-7B
Metrics	accuracy, precision, recall, F1, ROC-AUC, plus wall-clock latency per clip with 95% confidence intervals

ARSI works frame by frame, differencing then cropping then querying the model, so comparing our F1 to theirs without matching the unit of evaluation would be indefensible. Better to make it its own experiment, something like "compatibility with an established VLM anomaly-detection protocol", where ARSI is deliberately adapted to their evaluation unit. That gives a strong ablation:

METHOD	ACC	PRECISION	RECALL	F1	ROC-AUC	LATENCY
Qwen2.5-VL, from Borodin et al.						
InternVL3, from Borodin et al.						
ARSI, VLM only						
ARSI, differencing + VLM						

The question it answers is how much the change-detection front end really adds over the model on its own. Note that on UCF-Crime the differencing stage is degraded or switched off entirely, because there is no reference frame to subtract. Say so explicitly. "Our pipeline reduces to X when no reference is available" reads as a characterisation of the method, not as a weakness.

The same two-by-two comparison (off-the-shelf versus fine-tuned, full frame versus differencing) appears in [LORA PLAN.md](#) under "An open research question". The fine-tuning work is what produces the cells this table needs.

Gaps worth knowing about

Graffiti has no serious academic benchmark. There is a roughly 707-image annotated set on Roboflow and a handful of YOLOv8 conference papers, but nothing citable at the level of a vision venue. UCF-Crime does have a `Vandalism` class whose definition mentions graffiti and defacement, but it frames the problem differently in kind:

UCF-Crime	a person commits vandalism	→	an event in a video
ARSI	clean wall → wall with graffiti	→	a persistent change of state

So it is not a substitute for the Ostrava data. That gap is an argument *for* our dataset rather than a weakness in it, and it should be stated explicitly in the paper.

If the framing later shifts from producing a detection score to having the model explain the anomaly, the relevant benchmarks are UCA (23 542 sentence-level annotations over 1854 UCF-Crime videos), CUVA and its successor ECVA, HAWK, VAU-Bench and Holmes-VAU. Given how the pipeline currently works, differencing then cropping then judging the crop pair, the fixed-camera detection axis is the better fit.

Metric conventions to respect

Each community reports different numbers, and using the wrong ones makes results look non-comparable even when they are not.

- Change detection (CDnet 2014): F-measure per category, alongside the usual table of recall, precision, false-positive rate, false-negative rate and percentage of wrong classifications.
- Abandoned-object detection (AVSS, PETS, ABODA): precision, recall and F1 at the level of events, following the Luna et al. protocol.
- Clip-level vision-language models (UCF-Crime, Borodin protocol): accuracy, precision, recall, F1 and ROC-AUC, plus latency per clip with 95% confidence intervals.
- Semi-supervised fixed-camera anomaly detection, should we ever add a classic row such as ShanghaiTech: area under the ROC curve computed per frame.

